FED4FED: Statistical Fairness Evaluation Report

Privacy-Preserving Verification for Federated Learning Models

FAIRNESS VERDICT: UNFAIR (Statistical Disparity Detected)

Under the Conservative Joint Policy (Disparity detected if either Alg 1 or Alg 2 rejects H0), the global federated model exhibits statistically significant performance disparities across clients at alpha = 0.05. Alg 1 p-value: 4.9106e-02, Alg 2 p-value: 5.8555e-02.

1. Experiment & Architecture Configuration

Experiment ID:	mnist_fair_20261012_234829	**Date / Time:**	2026-10-12 23:50:00
Dataset:	MNIST	**Scenario:**	Fair
Clients (K):	100	**Seed:**	42
FL Rounds:	5	**Final Val Accuracy:**	0.8236
Partition Method:	uniform	**Bootstrap Replications (B):**	100

2. Statistical Fairness Hypothesis Tests

Test Algorithm	Statistic	df	p-value	Significance (α)	Decision
Algorithm 1 (Graybill-Deal / T1)	123.3673	99	0.0491	0.050	**Reject H0 (Disparity)**
Algorithm 2 (Welch-Type F)	1.2342	(99, 6767.3)	0.0586	0.050	**Reject H0 (Disparity)**

3. Algorithm 3: Global Performance Confidence Interval

Status: UNAVAILABLE (Condition Not Met)
Decision Rule Rationale: Algorithm 2 (Welch-Type F Test) rejected the null hypothesis of equal performance (p = 5.8555e-02 < alpha = 0.05). Because performance disparity is statistically significant across clients, the common mean assumption mu_1 = ... = mu_K does not hold. A pooled global confidence interval cannot be constructed under Algorithm 3 decision rules.

4. Sample Decentralized Client Performance Statistics

Client ID	Test Samples (n_i)	Observed Metric	Bootstrap Mean	Variance (S_i^2)	Std Error (SE)
Client 00	100	0.8600	0.8600	0.000891	0.0298
Client 01	100	0.7700	0.7644	0.001494	0.0386
Client 02	100	0.8200	0.8186	0.001685	0.0410

Client 03	100	0.8100	0.8067	0.001651	0.0406
Client 04	100	0.8200	0.8159	0.001647	0.0406
Client 05	100	0.8300	0.8236	0.001561	0.0395
Client 06	100	0.8400	0.8420	0.001772	0.0421
Client 07	100	0.7200	0.7138	0.002008	0.0448

Scientific Methodology: Fed4Fed privacy-preserving statistical homogeneity testing using non-parametric bootstrapping, Graybill-Deal common mean estimation, and Welch-type heteroscedastic F-testing. Generated by FED4FED Research Suite v1.0.0.